

Network Position and Academic Performance

Adding Network Centrality to Smirnov & Thurner (2017)

ECC3841 Network Economics — Research Brief

Introduction

A student's grades depend on effort and ability. They also depend on the student's friends. A friendship network gives access to information, advice, study partners, and motivation. So where a student sits in the network may matter for grades. We ask whether it does.

Research question

Does a student's position in a friendship network affect their grades, on top of who their direct friends are?

Background

Two papers frame the project.

Smirnov and Thurner (2017) tracked a changing “who-likes-who” network of Russian students. They found that the link between friends' grades is mostly *selection*: students pick friends with similar GPA, rather than being pulled up or down by them. Their method looks at one number per student — the average GPA of that student's direct friends. It ignores the rest of the network.

Calvó-Armengol, Patacchini and Zenou (2009; IZA DP 3859) supply the missing part. In their model, a student's effort is proportional to their *Katz–Bonacich centrality*, a measure of network position. On US data, more central students do better in school. But their data are one snapshot in time. So they cannot tell whether position raises grades, or good grades just attract more friends.

We combine the two. We take the centrality prediction to Smirnov and Thurner's data, which covers many time points, and test it while holding friend-group GPA and past GPA fixed.

The model

Each student i picks study effort x_i . Their payoff is

$$u_i = a x_i - \frac{1}{2} x_i^2 + \phi \sum_j g_{ij} x_i x_j.$$

The first term is the gain from effort. The second is its rising cost. The third is the social part: $g_{ij} = 1$ if i and j are friends, and $\phi > 0$ means a friend studying more raises i 's own reward from studying.

Setting the derivative to zero gives each student's best choice:

$$x_i = a + \phi \sum_j g_{ij} x_j.$$

Study a baseline amount, plus more when your friends study more. Solving for everyone at once:

$$\mathbf{x} = a(\mathbf{I} - \phi \mathbf{G})^{-1} \mathbf{1} = a \cdot \mathbf{b}(g, \phi).$$

So effort — and grades — track $\mathbf{b}(g, \phi)$, the Katz–Bonacich centrality. It expands as

$$b_i = 1 + \phi (\text{direct friends}) + \phi^2 (\text{friends of friends}) + \phi^3 (\dots) + \dots$$

a count of everyone you can reach by paths of any length, with longer paths weighted less.

What ϕ controls

ϕ sets how far influence spreads through the network.

- Small ϕ : only direct friends count. Centrality is almost the same as *popularity* (in-degree).
- Large ϕ (near its ceiling): distant links count almost as much as close ones. Centrality now measures *reach* — being a bridge between many groups.

So ϕ is a dial from “popularity” to “reach.” We compute centrality at several values of ϕ and see how the result changes along the dial.

The ceiling on ϕ is $1/\lambda_{\max}$, where $\lambda_{\max}$ is the largest eigenvalue of $\mathbf{G}$; above it the model has no solution. $\lambda_{\max}$ is larger in bigger, denser networks, so the ceiling differs across our 38 networks. We set $\phi = c \cdot (1/\lambda_{\max})$ with c from 0.05 to 0.95. Then c means the same thing in every network, and results can be compared.

Empirical strategy

For each network we compute centrality at each c , then run

$$\begin{aligned} \text{GPA}_{i,t+1} = & \beta_0 + \beta_1 \text{Katz}_{i,t} + \beta_2 \text{GPA}_{i,t} + \beta_3 \overline{\text{GPA}}_{i,t}^{\text{friends}} \\ & + \beta_4 \text{InDeg}_{i,t} + \beta_5 \text{OutDeg}_{i,t} + \gamma_{\text{group}} + \varepsilon_{i,t}. \end{aligned}$$

Each control has a job. Average friend GPA is Smirnov and Thurner’s variable, so β_1 is net of friend-group composition. In- and out-degree are raw popularity, so β_1 is net of that too. The real question is whether centrality adds anything beyond counting friends. We run this at every c . One test, at $c = 0.85$, is fixed in advance as the headline. We also try betweenness and eigenvector centrality.

Which way does the arrow run?

Does position raise grades, or do grades attract friends? One snapshot cannot tell. Our data help in three ways:

1. **Time order.** Position is measured before the GPA it predicts.
2. **Past GPA is controlled.** We compare students who currently have the same grade.
3. **Panel data.** We can remove fixed traits (ability, background) with student fixed effects, and we can measure the reverse arrow directly: does a rise in GPA bring new friends?

What we still cannot rule out: something that changes a student mid-term and lifts both grades and friend count. “Likes” are also a rough stand-in for real friendship. So our claim is that position *predicts* later grades, not that it *causes* them.

Predictions

From the model: $\beta_1 > 0$, and centrality beats raw degree. The claim fails if β_1 is not significant once degree is controlled, or if it flips sign along the ϕ dial.

Reading the sign of β_1 :

- $\beta_1 > 0$: central students gain from more access to information and help.
- $\beta_1 = 0$: the gains and the time costs cancel.
- $\beta_1 < 0$: wide reach costs study time.

From replicating Smirnov and Thurner:

- A friend’s past GPA does not predict a student’s future GPA, once own past GPA is controlled.
- New friendships show a smaller GPA gap than ending ones — a sign of selection.
- Selection is stronger at university than in high school, where students can change their circle more easily.

Data and network

Data: Smirnov & Thurner (2017) replication files, Harvard Dataverse (doi:10.7910/DVN/SZA9YW). About 6,200 Russian students, grades plus “like” ties every 3 months.

Nodes: students. 655 in high school; 1,200–1,549 per university year.

Edges: directed “like” ties. $i \rightarrow j$ means i liked j at least once in a 3-month window. 2 to 14 snapshots per group, 38 in total. Only 24–27% of ties go both ways.

Attributes: GPA; in-degree, out-degree, Katz–Bonacich (several ϕ), betweenness, eigenvector; high school vs. university and year.

Related studies

1. Calvó-Armengol, Patacchini & Zenou (2009), “Peer Effects and Social Networks in Education” (IZA DP 3859). The model above, tested on AddHealth friendship data. Limit: one snapshot, so cause and effect are not separated.
2. Giulietti, Vlassopoulos & Zenou (2020), “Peers, Gender, and Long-Term Depression” (IZA DP 13680). Peer depression in school predicts adult depression, for women. Uses reference groups, not friendship ties. Zenou teaches this course.
3. Smirnov & Thurner (2017), *PLOS ONE* 12(8): e0183473. The paper and data this project replicates and extends.